1st Sept 2026

<u>To Whom It May Concern</u>

Project: Al Joud Tower-B+G+5Parking+Service floor+48 floors +Roof
Client: Al Batha Real Estate
Consultant: Cubic
Main Contractor: Al Wathba
Mep Contractor: GECO Mechanical & Electrical Ltd. Co.

Subject: Technical details for Fusion Bonded Epoxy (FBE) Coating on Iron Valves

We hereby submit this letter to clarify the concept and application of Fusion Bonded Epoxy (FBE) coating for iron-bodied valves proposed for the project.

Fusion Bonded Epoxy is a thermoset epoxy powder coating applied to the prepared surface of the iron valve body through a controlled heat-fusion process. The coating forms a continuous, strongly bonded and corrosion-resistant protective layer over the iron substrate.

Purpose of FBE Coating

The primary purpose of applying FBE coating to iron valves is to provide effective protection against:

- Corrosion and oxidation of the iron valve body.
- Moisture and water exposure.
- External environmental conditions.
- Deterioration during storage, transportation and installation.
- Potential corrosion in buried or otherwise corrosive service environments, where applicable.

FBE Coating Process

The valve body is first subjected to appropriate surface preparation to remove contaminants. The prepared iron surface is then heated to the specified temperature, after which the epoxy powder is electrostatically applied. The powder melts and chemically cures on the heated surface, forming a uniform and durable coating.

This process provides a strong bond between the epoxy coating and the iron substrate and minimizes the risk of coating separation under normal service conditions.

Advantages of FBE Coating

FBE coating provides several advantages for iron valves, including:

1. Excellent corrosion protection of the iron substrate.
2. Strong adhesion to the prepared metal surface.
3. Good resistance to water and moisture.
4. High durability and mechanical resistance compared with conventional liquid paint systems.
5. Uniform and continuous coating, including protection of the valve body surfaces.

6. Good resistance to a wide range of chemicals and environmental conditions, subject to the service conditions and coating specification.
7. Long-term protection when properly applied and maintained.

The FBE coating therefore acts as a protective barrier between the iron valve body and the surrounding environment, significantly reducing the possibility of corrosion and extending the service life of the valve.

Yours faithfully,

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